

Simultaneous State and Disturbances Estimation for Sensorless PMSM Control: Design and Experiments

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Abstract—Sensorless control for permanent magnet synchronous motor (PMSM) is drawing increasing research interest in both academia and industry. To enhance the robustness of sensorless control schemes, disturbance observers are typically designed for feedforward disturbance compensation. However, the conventional framework, which separately estimates speed/position and disturbances, may lead to system instability, particularly when disturbances significantly affect speed/position estimation performance. To address this issue, this article proposes a novel estimation-based control framework that simultaneously estimates system states and disturbances. Within this framework, two simultaneous observers are introduced. First, an augmented extended Kalman filter that extends disturbances as additional states and second, a composite extended Kalman filter integrating disturbance observer and extended Kalman filter. The feasibility and effectiveness of the proposed sensorless control schemes are validated through extensive comparative hardware experiments. Compared with the conventional separate-estimation framework, the proposed approach significantly improves speed and position-estimation accuracy under load disturbances in sensorless PMSM drives.

Index Terms—Disturbance observer (DO), extended Kalman filter (EKF), permanent magnet synchronous motor (PMSM), sensorless control, simultaneous estimation.

I. INTRODUCTION

OVER the past decades, sensorless motor control has attracted significant attention from both academic researchers and industrial practitioners. This interest stems from

the technique's notable application potentials in system implementation and reliability. Specifically, sensorless control leverages electrical variables (e.g., currents and voltages) to estimate motor states (i.e., position and speed) instead of relying on mechanical sensors (e.g., optical encoders, resolvers, and tacho-generators). This approach offers advantages, such as cost reduction, system simplification, enhanced reliability, and adaptability in diverse application environments [1], [2].

To date, many sensorless control schemes have been proposed, which can be divided into two main categories including signal injection schemes and fundamental-model-based schemes [3]. Signal injection schemes offer superior estimation performance in the zero-speed and low-speed domains. They leverage the spatial information of existing anisotropies excited by injected voltage or current signals for rotor position estimation [4]. However, these schemes introduce additional power losses and acoustic noises [5]. While fundamental-model-based schemes are better suited for the medium-speed and high-speed domains. They typically utilize back electromotive forces (BEMFs) or flux linkages (integral of the BEMFs) as an intermediate variable to further estimate rotor position and speed. According to the observer/filter used for estimation, this class of sensorless schemes can be further broadly classified into categories, such as extended Kalman filter (EKF) [6], [7], sliding mode observer (SMO) [8], model reference adaptive [9], and extended state observer [10], [11]. Each approach achieves satisfactory estimation performance for sensorless control from different perspectives.

However, it should be noted that most sensorless schemes are designed based on a relatively ideal motor systems without considering external disturbance or load torque. However, disturbances and load variations are widespread and inevitable in real-life applications and can significantly degrade motor system performance [12]. In particular, under heavy-load conditions, disturbances can severely impact position and speed-estimation accuracy. Moreover, since these estimations are directly used in feedback control, excessive estimation errors may compromise system stability. Therefore, achieving accurate position and speed estimation is crucial for ensuring both system stability and sensorless control performance.

An intuitive idea is to construct disturbance observer (DO) to estimate the disturbance/load based on measurable variables [13]. In this context, in [14], a separate DO and a saliency-tracking observer are jointly designed to enhance disturbance rejection performance. In [15], a sensorless control scheme employing two cascaded observers, i.e., an SMO for speed/position

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estimation and a DO for load torque estimation, is proposed for permanent magnet synchronous motor (PMSM) drive. To further enhance the capability of SMO in suppressing load disturbance, Jin et al. [16] introduced a DO based on integral sliding mode control. Notably, in these sensorless schemes, the speed/position observer and DO are designed separately. However, this control framework, which estimates system states and disturbances independently, may lead to system instability, particularly when disturbances significantly degrade speed/position estimation accuracy.

To this end, this article presents a simultaneous estimation control framework for sensorless control, i.e., simultaneous estimation of system states and disturbances. In this context, Janiszewski [17] proposed an augmented extended Kalman filter (AEKF) to achieve simultaneous estimation by treating external disturbance as an extended state. However, this approach inevitably increases the computational burden and may lead to the unobservability issue. To this end, this article proposes an alternative approach for achieving simultaneous states and disturbance estimation through a composite design of DO and state observer. Both estimated disturbances are employed in feedforward compensation to effectively mitigate their adverse effects. For clarity, the main contributions of this study are summarized as follows.

- 1) A simultaneous estimation control framework is introduced for sensorless PMSM control under unknown disturbances. Theoretical analysis and comparative results validate the effectiveness of this control framework.
- 2) A composite EKF (CEKF) based on DO is developed for simultaneous estimation of motor states and unknown load disturbances. The DO integrated within CEKF plays the role of real-time disturbance estimation and state update. Comparative experiments show that the proposed CEKF performs better than conventional EKF and AEKF in terms of speed/position estimation and disturbance rejection capability.
- 3) The proposed state-disturbance simultaneous estimation control framework can be applied to a wide range of disturbance rejection control problems beyond sensorless PMSM control. Besides, the proposed CEKF can also be applied to different motor drive systems with minimal modification.

The rest of this article is organized as follows. Section II formulates the problem under consideration. The main theoretical results are presented in Section III, which are applied in sensorless PMSM control in Section IV. Extensive experimental validations are conducted in Section V. Finally, Section VI concludes this article with future work.

II. PROBLEM AND PRELIMINARIES

A. Problem Formulation

Real-time estimation of system states and disturbances is commonly employed in various engineering systems, especially for sensorless control of motor systems [2]. To illustrate the estimation and control problem in sensorless techniques, we

consider the following disturbed double integrator system:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = u + d(t) \\ y = x_1 \end{cases} \quad (1)$$

where $x = [x_1, x_2]^T$ is the system state, u is the control input, y is the system output, and $d(t)$ represents an external unknown disturbance (i.e., unknown load).

The objective of this article is to design an observer to simultaneously estimate unmeasurable state x_2 and unknown disturbance $d(t)$ by using y and u along with system dynamics.

Assumption 1: The disturbance $d(t)$ satisfies the following conditions: 1) $d(t)$ and its derivative are bounded, i.e., $|d(t)| \leq d^*$ and $|\dot{d}(t)| \leq D^*$; 2) disturbance $d(t)$ has a constant value in steady state, i.e., $\lim_{t \rightarrow \infty} \dot{d}(t) = 0$ and $\lim_{t \rightarrow \infty} d(t) = D$.

B. Useful Lemmas

Lemma 1 (see [18]): Consider the following system:

$$\dot{x} = f(t, x, u) \quad (2)$$

where $f : [0, \infty) \times R^n \times R^m \rightarrow R^n$ is continuously differentiable and globally Lipschitz in x and u , uniformly in t . If the unforced system $\dot{x} = f(t, x, u)$ has a globally exponentially stable equilibrium point at the origin $x = 0$, then the system $\dot{x} = f(t, x, u)$ is input-to-state stable (ISS).

Lemma 2 (see [19]): If the system (2) is ISS, then the system states satisfy $\lim_{t \rightarrow \infty} x(t) = 0$ when $\lim_{t \rightarrow \infty} u(t) = 0$.

C. Controller Design

Assume that the desired trajectory $y_d(t)$ and its derivatives $\dot{y}_d(t)$ and $\ddot{y}_d(t)$ are known functions. The PID controller with disturbance compensation is designed as

$$\begin{aligned} u = & -k_p(x_1 - y_d) - k_i \int_0^t (x_1 - y_d) d\tau \\ & - k_d(\hat{x}_2 - \dot{y}_d) - \hat{d} + \ddot{y}_d \end{aligned} \quad (3)$$

where $k_p > 0$, $k_i > 0$, and $k_d > 0$ are proper proportional, integral, and differential coefficients, respectively, and $\hat{x}_2$ and $\hat{d}$ are state and disturbance information estimated from the designed observer in the following section.

III. THEORETICAL RESULTS

This section presents the theoretical results. First, the conventional separate estimation scheme is introduced in Section III-A, followed by the proposed simultaneous estimation scheme in Section III-B.

A. Conventional Separate Estimation Control Scheme

The conventional separate estimation scheme is shown in Fig. 1, which includes a Luenberger state observer

$$\begin{cases} \dot{\hat{x}}_1 = \hat{x}_2 + l_1(x_1 - \hat{x}_1) \\ \dot{\hat{x}}_2 = u + l_2(x_1 - \hat{x}_1) \end{cases} \quad (4)$$

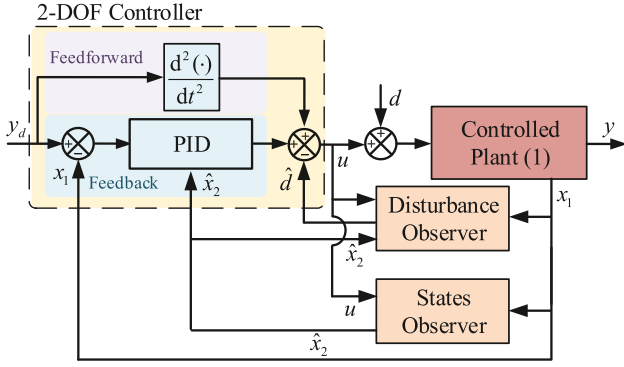


Fig. 1. Block diagram of conventional separate estimation control scheme.

and a linear DO in the form of

$$\begin{cases} \dot{p} = -l_3(\hat{d} + u) \\ \dot{\hat{d}} = p + l_3\hat{x}_2 \end{cases} \quad (5)$$

where $\hat{x} = [\hat{x}_1, \hat{x}_2]^T$ is the estimated state, p is the internal state, $\hat{d}$ is the estimated disturbance, and $l_i > 0, i = 1, 2, 3$ are constant observer gains to be designed.

The stability analysis of system (1) with observers (4) and (5) is given as follows. The estimation errors of observer (4) are denoted by $e_1 = x_1 - \hat{x}_1, e_2 = x_2 - \hat{x}_2$, and define $e = [e_1, e_2]^T$, then the dynamics of e is given by

$$\begin{aligned} \dot{e} &= \underbrace{\begin{bmatrix} -l_1 & 1 \\ -l_2 & 0 \end{bmatrix}}_{A_e} e + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B_e} d \\ &= A_e e + B_e d. \end{aligned} \quad (6)$$

By choosing the gains $l_1 > 0$ and $l_2 > 0$, it is easy to verify that A_e is a Hurwitz matrix. According to Lemma 1, it follows that the closed loop system (6) with d as input is ISS. From system (6), it is obtained that $\ddot{e}_1 = -l_1\dot{e}_1 + \dot{e}_2$ and $\ddot{e}_2 = -l_2\dot{e}_1 + \dot{d}$. Combining these two equations, one has

$$\ddot{e}_1 + l_1\dot{e}_1 + l_2\dot{e}_2 = \dot{d}. \quad (7)$$

It follows from Lemma 1 that system (7) with $\dot{d}$ as input is ISS. Under Assumption 1, it follows from Lemma 2 that the system states can converge to zero, i.e., $\lim_{t \rightarrow \infty} \dot{e}_1(t) = 0$ and $\lim_{t \rightarrow \infty} \ddot{e}_1(t) = 0$. Since $\ddot{e}_1 = -l_1\dot{e}_1 + \dot{e}_2$, it is obtained that

$$\lim_{t \rightarrow \infty} \dot{e}_2(t) = 0. \quad (8)$$

Let $e_3 = d - \hat{d}$. Based on (4) and (5), we have

$$\begin{aligned} \dot{e}_3 &= \dot{d} - (\dot{p} + l_3\dot{\hat{x}}_2) \\ &= \dot{d} + l_3(\hat{d} + u) - l_3(u + l_2e_1) \\ &= \dot{d} + l_3[(\hat{d} - d) + (d - l_2e_1)]. \end{aligned} \quad (9)$$

From system (6), one has $\dot{e}_2 = -l_2e_1 + d$. Substituting it into system (9), it is obtained that

$$\dot{e}_3 = \dot{d} - l_3e_3 - l_3\dot{e}_2. \quad (10)$$

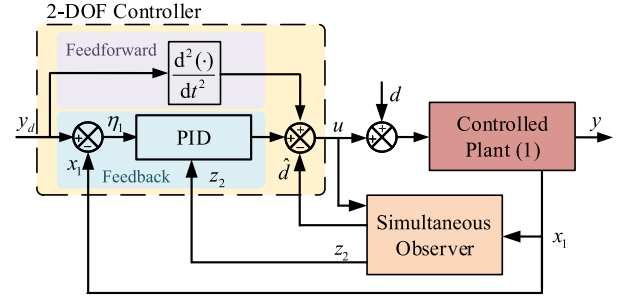


Fig. 2. Block diagram of simultaneous estimation control scheme.

By Assumption 1, it is easy to verify that system (10) with $-l_3\dot{e}_2$ as input is ISS. Because the origin of e_2 -dynamics is globally uniformly asymptotically stable, using criterion of ISS in the cascade system, it can be concluded that the origin of system (10) is asymptotically stable, that is

$$\lim_{t \rightarrow \infty} \dot{e}_3(t) = 0. \quad (11)$$

Thus, the states and disturbance estimation errors satisfy: $\lim_{t \rightarrow \infty} \dot{e}_i(t) = 0, i = 1, 2, 3$.

Next, the stability analysis of the closed-loop control systems (1) and (3) is given as follows. Denote the reference signal tracking errors as $\eta_1 = x_1 - y_d$ and $\eta_2 = x_2 - \dot{y}_d$ and define $\eta = [\eta_1, \eta_2]^T$. The dynamics of η can be expressed as

$$\begin{cases} \dot{\eta}_1 = \eta_2 \\ \dot{\eta}_2 = -k_p\eta_1 - k_i \int_0^t \eta_1 d\tau - k_d(\eta_2 - e_2) + e_3. \end{cases} \quad (12)$$

Taking the derivative of $\dot{\eta}_2$, it follows that

$$\ddot{\eta}_1 + k_p\dot{\eta}_1 + k_i\eta_1 + k_d\ddot{\eta}_1 = k_d\dot{e}_2 + \dot{e}_3. \quad (13)$$

The control gains k_p, k_i , and k_d are selected such that $\lambda(s) = s^3 + k_d s^2 + k_p s + k_i$ is a Hurwitz polynomial. Thus, the left-hand side of (13) is globally asymptotically stable system. Based on the stability analysis of observer system, it is obtained that $k_d\dot{e}_2 + \dot{e}_3$ is bounded and $\lim_{t \rightarrow \infty} [k_d\dot{e}_2(t) + \dot{e}_3(t)] = 0$. According to Lemma 1 and criterion of ISS in the cascade system, it follows that the closed-loop system (13) is ISS and the system states will asymptotically converge to zero, that is

$$\begin{cases} \lim_{t \rightarrow \infty} \eta_1(t) = 0 \\ \lim_{t \rightarrow \infty} \eta_2(t) = 0. \end{cases} \quad (14)$$

This completes the proof.

B. Simultaneous Estimation Control Scheme

To enhance estimation performance, this section proposes a simultaneous estimation scheme for jointly estimating states and disturbances, as illustrated in Fig. 2. The proposed simultaneous observer is formulated as follows:

$$\begin{cases} \dot{z}_1 = z_2 + h_1(x_1 - z_1) \\ \dot{z}_2 = z_3 + h_2(x_1 - z_1) \\ \dot{z}_3 = h_3(x_1 - z_1) \end{cases} \quad (15)$$

where $[z_1, z_2]^T$ are the estimated values of states $[x_1, x_2]^T$, z_3 is the estimated value of external disturbance, and $h_i > 0, i =$

1, 2, 3 are coefficients of the Hurwitz polynomial $\lambda(s) = s^3 + h_1s^2 + h_2s + h_3$.

The stability analysis of system (1) with observer (15) is given as follows. Define the estimation errors as

$$\begin{cases} \varepsilon_1 = x_1 - z_1 \\ \varepsilon_2 = x_2 - z_2 \\ \varepsilon_3 = d - z_3. \end{cases} \quad (16)$$

Let $\varepsilon = [\varepsilon_1, \varepsilon_2, \varepsilon_3]^T$, then the dynamics of ε is expressed as

$$\begin{aligned} \dot{\varepsilon} &= \underbrace{\begin{bmatrix} -h_1 & 1 & 0 \\ -h_2 & 0 & 1 \\ -h_3 & 0 & 0 \end{bmatrix}}_{A_\varepsilon} \varepsilon + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{B_\varepsilon} \dot{d} \\ &= A_\varepsilon \varepsilon + B_\varepsilon \dot{d}. \end{aligned} \quad (17)$$

Since A_ε is a Hurwitz matrix, the origin of system $\dot{\varepsilon} = A_\varepsilon \varepsilon$ is globally asymptotically stable. Then, it follows from Lemma 1 that the system (17) with $\dot{d}$ as input is ISS. By Assumption 1 and Lemma 2, it is obtained that $\lim_{t \rightarrow \infty} \dot{\varepsilon}_i(t) = 0, i = 1, 2, 3$.

Next, the stability analysis of the closed loop control systems (1) and (3) is given as follows. Denote the tracking errors as

$$\begin{cases} \eta_1 = x_1 - y_d \\ \eta_2 = x_2 - \dot{y}_d. \end{cases} \quad (18)$$

Let $\eta = [\eta_1, \eta_2]^T$. The dynamics of η can be expressed as

$$\begin{cases} \dot{\eta}_1 = \eta_2 \\ \dot{\eta}_2 = -k_p \eta_1 - k_i \int_0^t \eta_1 d\tau - k_d(\eta_2 - \varepsilon_2) + \varepsilon_3. \end{cases} \quad (19)$$

Taking the derivative of $\dot{\eta}_2$, it follows that

$$\ddot{\eta}_1 + k_p \dot{\eta}_1 + k_i \eta_1 + k_d \ddot{\eta}_1 = k_d \dot{\varepsilon}_2 + \dot{\varepsilon}_3. \quad (20)$$

The control gains k_p, k_i , and k_d are selected such that $\lambda(s) = s^3 + k_d s^2 + k_p s + k_i$ is Hurwitz. Thus, the left-hand side of (20) is a globally asymptotically stable system. Based on the stability analysis of observer system, it is obtained that $k_d \dot{\varepsilon}_2 + \dot{\varepsilon}_3$ is bounded and $\lim_{t \rightarrow \infty} [k_d \dot{\varepsilon}_2(t) + \dot{\varepsilon}_3(t)] = 0$. According to Lemma 1 and the criterion of ISS in the cascade system, it follows that the closed loop system (20) is ISS and the system states will asymptotically converge to zero, i.e.,

$$\begin{cases} \lim_{t \rightarrow \infty} \eta_1(t) = 0 \\ \lim_{t \rightarrow \infty} \eta_2(t) = 0. \end{cases} \quad (21)$$

This completes the proof.

The main difference between separate estimation control scheme and simultaneous estimation control scheme lie in state estimation. As shown in (6) and (17), the simultaneous estimation control scheme can cope with higher order disturbance, and maintain overall system stability. For instance, suppose load disturbance satisfies $\ddot{d} = 0$ and $\dot{d} = a \neq 0$ with a is an unknown constant. According to (6), the state estimation error $e(t)$ of the separate estimation control scheme diverges and remains unbounded. Then, the closed loop system (13) fails to achieve accurate trajectory tracking control. However, for the simultaneous estimation control scheme, state estimation error $\varepsilon(t)$ is ISS and bounded according to (17). Thus, the closed loop system (20) is still globally asymptotically stable.

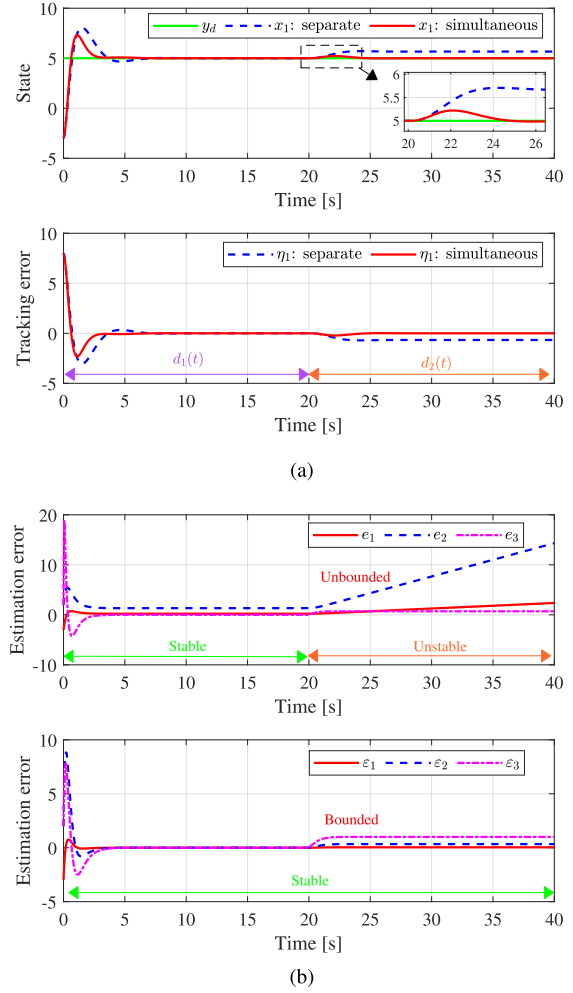


Fig. 3. Tracking performance of the desired trajectory y_d . (a) State tracking performance and state tracking errors. (b) State estimation errors: separate scheme (upper) and simultaneous scheme (lower).

C. Comparative Simulation Comparisons

The simulation comparisons are given in this section to demonstrate the effectiveness of these two schemes. To make a fair comparison, the controller parameters are taken as $k_p = 8$, $k_i = 6$, and $k_d = 6$. The disturbance is set as follows:

$$d(t) = \begin{cases} d_1(t) : 2 + e^{-t} \sin(t) & 0 \leq t \leq 20 \text{ s} \\ d_2(t) : t - 18 & t > 20 \text{ s}. \end{cases} \quad (22)$$

For a fair comparison, observer gains selection method in these two schemes is based on bandwidth-parameterization [21]. In the separate observer scheme, the Luenberger state observer gains are set as: $l_1 = 6$ and $l_2 = 9$, and the DO gain is set as $l_3 = 27$. In simultaneous estimation scheme, the gains of simultaneous observer are set as: $h_1 = 9$, $h_2 = 27$, and $h_3 = 27$. The desired trajectory is set as $y_d(t) = 5$.

Fig. 3 presents the tracking performance of the desired trajectory y_d under these two estimation control schemes. In Fig. 3(a), from top to bottom, the state tracking performance and state tracking errors are shown. Clearly, the simultaneous estimation control scheme demonstrates a faster convergence time than the

separate estimation control scheme under the same bandwidth. In Fig. 3(b), from top to bottom, the state estimation errors of the separate and simultaneous estimation control schemes are presented. As observed, the separate observer scheme exhibits a constant estimation error e_2 for the unmeasurable state x_2 in the steady state. This error arises because the unknown disturbances are not accounted for in the state observation process. When $t > 20$ s and the disturbance $\dot{d}(t) \neq 0$, only the simultaneous estimation control scheme can accurately track the desired trajectory, demonstrating superior disturbance rejection capability.

IV. APPLICATION IN SENSORLESS PMSM CONTROL

This section presents the application design. First, mathematical model of PMSM is introduced in Section IV-A, followed by two estimation schemes in Sections IV-B and IV-C. Then, composite speed controller for these sensorless schemes is shown in Section IV-D.

A. PMSM Mathematical Model

The mathematical model of surface PMSM (SPMSM) in the stationary $\alpha\beta$ reference frame can be expressed as

$$\begin{cases} \dot{i}_\alpha = -\frac{R_s}{L_s}i_\alpha + \frac{1}{L_s}u_\alpha + \frac{\omega_e\psi_f \sin \theta_e}{L_s} \\ \dot{i}_\beta = -\frac{R_s}{L_s}i_\beta + \frac{1}{L_s}u_\beta - \frac{\omega_e\psi_f \cos \theta_e}{L_s} \end{cases} \quad (23)$$

where i_α and i_β are the $\alpha\beta$ -axes phase current components, u_α and u_β are the $\alpha\beta$ -axes terminal voltage components, and e_α and e_β are the $\alpha\beta$ -axes BEMF components, respectively. R_s is the stator resistance, L_s is the stator inductance, ω_e is the electrical speed, ψ_f is the permanent-magnet flux linkage, and θ_e is the electrical angle.

The dynamic mechanical equation of SPMSM is given by

$$\begin{cases} \dot{\theta}_e = \omega_e \\ \dot{\omega}_e = K_T(i_\beta \cos \theta_e - i_\alpha \sin \theta_e) - \frac{n_p}{J}T_L - \frac{B}{J}\omega_e \end{cases} \quad (24)$$

where $K_T = \frac{3\psi_f n_p^2}{2J}$ is the torque coefficient, n_p is the number of pole pair, J is the moment of inertia, B is the viscous friction coefficient, and T_L is the load torque.

Then, the state space model of SPMSM according to (23) and (24) can be expressed as

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{B}_u \mathbf{u} \\ \mathbf{y} = \mathbf{h}(\mathbf{x}) \end{cases} \quad (25)$$

where $\mathbf{x} = [i_\alpha \ i_\beta \ \omega_e \ \theta_e]^T$ is the state variant, $\mathbf{u} = [u_\alpha \ u_\beta]^T$ is the control input, $\mathbf{y} = [i_\alpha \ i_\beta]^T$ is the measurable output, the system functions or matrices $\mathbf{f}(\mathbf{x})$, $\mathbf{B}_u$, and $\mathbf{h}(\mathbf{x})$ are

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix} = \begin{bmatrix} -\frac{R_s}{L_s}i_\alpha + \frac{\omega_e\psi_f \sin \theta_e}{L_s} \\ -\frac{R_s}{L_s}i_\beta - \frac{\psi_f\omega_e \cos \theta_e}{L_s} \\ K_T(i_\beta \cos \theta_e - i_\alpha \sin \theta_e) - \frac{n_p}{J}T_L - \frac{B}{J}\omega_e \\ \omega_e \end{bmatrix}$$

$$\mathbf{B}_u = \begin{bmatrix} \frac{1}{L_s} & 0 \\ 0 & \frac{1}{L_s} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \mathbf{h}(\mathbf{x}) = \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}. \quad (26)$$

It should be noted that among mainstream sensorless control schemes for PMSM drives [6], [7], [8], [9], [10], [11], EKF is the only approach that utilizes both the current-voltage mathematical (23) and the dynamic mechanical (24) for motor state estimation. Moreover, common external disturbances (e.g., load disturbances) are explicitly represented in the dynamic mechanical equations. Thus, this article adopts EKF as the specific algorithm to implement the proposed theoretical framework. In addition, the load disturbance T_L is regarded as the only disturbance d to be considered.

B. AEKF for Simultaneous Estimation (Case I)

Assumption 2. [17]: Assuming that the estimated variable T_L is constant in small interval (sampling time T_s), i.e., $\dot{T}_L = 0$.

Then, the SPMSM system (24) can be transformed to

$$\begin{cases} \dot{\bar{\mathbf{x}}} = \bar{\mathbf{f}}(\bar{\mathbf{x}}) + \bar{\mathbf{B}}_u \mathbf{u} + \boldsymbol{\omega} \\ \mathbf{y} = \bar{\mathbf{h}}(\bar{\mathbf{x}}) + \mathbf{v} \end{cases}$$

$$\bar{\mathbf{x}} = \begin{bmatrix} \mathbf{x} \\ T_L \end{bmatrix}, \bar{\mathbf{f}}(\bar{\mathbf{x}}) = \begin{bmatrix} \mathbf{f}(\mathbf{x}) \\ 0 \end{bmatrix}, \bar{\mathbf{B}}_u = \begin{bmatrix} (\mathbf{B}_u)_{4 \times 2} \\ \mathbf{0}_{1 \times 2} \end{bmatrix}, \bar{\mathbf{h}}(\bar{\mathbf{x}}) = \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}. \quad (27)$$

where $\boldsymbol{\omega}$ and $\mathbf{v}$ are zero-mean white Gaussian noises with uncorrelated covariance matrices $\mathbf{Q}$ and $\mathbf{R}$, respectively.

After the first-order forward-Euler approximation [20], the SPMSM model (27) can be expressed as

$$\begin{cases} \bar{\mathbf{x}}_k = \bar{\mathbf{x}}_{k-1} + T_s [\bar{\mathbf{f}}(\bar{\mathbf{x}}_{k-1}) + \bar{\mathbf{B}}_u \mathbf{u}_{k-1}] + \boldsymbol{\omega}_{k-1} \\ \mathbf{y}_k = \bar{\mathbf{h}}(\bar{\mathbf{x}}_k) + \mathbf{v}_k \end{cases} \quad (28)$$

where T_s is the sampling time.

Given the nonlinear nature of the system under consideration, AEKF is employed for the estimation/observer task. The AEKF computational process consists of two main steps. The first step is the prediction step, where a simple rectangular integration technique is applied, leading to the following recursive equations:

$$\begin{cases} \hat{\bar{\mathbf{x}}}_k^- = \hat{\bar{\mathbf{x}}}_{k-1} + T_s [\bar{\mathbf{f}}(\hat{\bar{\mathbf{x}}}_{k-1}) + \bar{\mathbf{B}}_u \mathbf{u}_{k-1}] \\ \mathbf{P}_k^- = (\mathbf{I} + T_s \bar{\mathbf{F}}_{k-1}) \mathbf{P}_{k-1} (\mathbf{I} + T_s \bar{\mathbf{F}}_{k-1})^T + \mathbf{Q}. \end{cases} \quad (29)$$

where $\bar{\mathbf{F}}_{k-1} = \frac{\partial \bar{\mathbf{f}}}{\partial \bar{\mathbf{x}}} |_{\bar{\mathbf{x}}=\hat{\bar{\mathbf{x}}}_{k-1}}$ and $\mathbf{P}$ is the covariance matrix.

The second step is the update step, which corrects the predicted state estimate and its covariance matrix. This is realized by the following recursive relations:

$$\begin{cases} K_k = \frac{\mathbf{P}_k^- \bar{\mathbf{H}}_k^T}{\bar{\mathbf{H}}_k \mathbf{P}_k^- \bar{\mathbf{H}}_k^T + \mathbf{R}_k} \\ \hat{\bar{\mathbf{x}}}_k = \hat{\bar{\mathbf{x}}}_k^- + K_k (\mathbf{y}_k - \mathbf{h}(\hat{\bar{\mathbf{x}}}_k^-)) \\ \mathbf{P}_k = (\mathbf{I} - K_k \bar{\mathbf{H}}_k) \mathbf{P}_k^- \end{cases} \quad (30)$$

where K_k is the Kalman gain, $\bar{\mathbf{H}}_k = \frac{\partial \bar{\mathbf{h}}}{\partial \bar{\mathbf{x}}} |_{\bar{\mathbf{x}}=\hat{\bar{\mathbf{x}}}_k}$ is the output transformation matrix, and the detailed calculation of $\bar{\mathbf{H}}_k$ is described in the appendix.

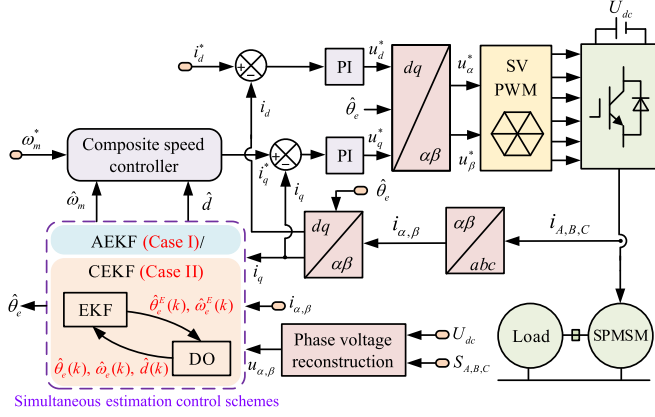


Fig. 4. Diagram of simultaneous estimation control for sensorless control.

Fig. 4 presents block diagram of the simultaneous estimation control framework for sensorless PMSM control. It contains two types of different simultaneous estimation control schemes, i.e., AEKF and CEKF (more details are shown in Section IV-C). Both AEKF and CEKF can independently and simultaneously estimate position, speed, and load disturbances.

C. CEKF for Simultaneous Estimation (Case II)

In Fig. 4, the CEKF integrates a DO into the EKF, enabling real-time estimation of load torque disturbances and state re-estimation. The detailed derivations are summarized in the following three steps.

1) *State Estimation With EKF*: Temporarily assuming the load torque disturbance T_L in $\mathbf{f}(\mathbf{x})$ is known, the SPMSM model (25) can be expressed after first-order forward-Euler approximation as

$$\begin{cases} \mathbf{x}_k = \mathbf{x}_{k-1} + T_s [\mathbf{f}(\mathbf{x}_{k-1}) + \mathbf{B}_u \mathbf{u}_{k-1}] + \boldsymbol{\omega}_{k-1} \\ \mathbf{y}_k = \mathbf{h}(\mathbf{x}_k) + \mathbf{v}_k. \end{cases} \quad (31)$$

The computational process of the EKF proceeds with the following two steps. The first step is the prediction step, and a simple rectangular integration technique is used so that the prediction step results in the following recursive equations:

$$\begin{cases} \hat{\mathbf{x}}_k^- = \hat{\mathbf{x}}_{k-1} + T_s [\mathbf{f}(\hat{\mathbf{x}}_{k-1}) + \mathbf{B}_u \mathbf{u}_{k-1}] \\ \mathbf{P}_k^- = (\mathbf{I} + T_s \mathbf{F}_{k-1}) \mathbf{P}_{k-1} (\mathbf{I} + T_s \mathbf{F}_{k-1})^T + \mathbf{Q} \end{cases} \quad (32)$$

where $\mathbf{F}_{k-1} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}|_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}}$ is the system matrix, and the detailed calculation is described in the appendix.

The second step is the update step, which corrects the predicted state estimate and its covariance matrix. This is realized by the following recursive relations:

$$\begin{cases} K_k = \frac{\mathbf{P}_k^- \mathbf{H}_k^T}{\mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^T + \mathbf{R}_k} \\ \hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + K_k [\mathbf{y}_k - \mathbf{h}(\hat{\mathbf{x}}_k^-)] \\ \mathbf{P}_k = (\mathbf{I} - K_k \mathbf{H}_k) \mathbf{P}_k^- \end{cases} \quad (33)$$

where $\mathbf{H}_k = \frac{\partial \mathbf{h}}{\partial \mathbf{x}}|_{\mathbf{x}=\hat{\mathbf{x}}_k}$ is the output transformation matrix, and the detailed calculation is described in the appendix.

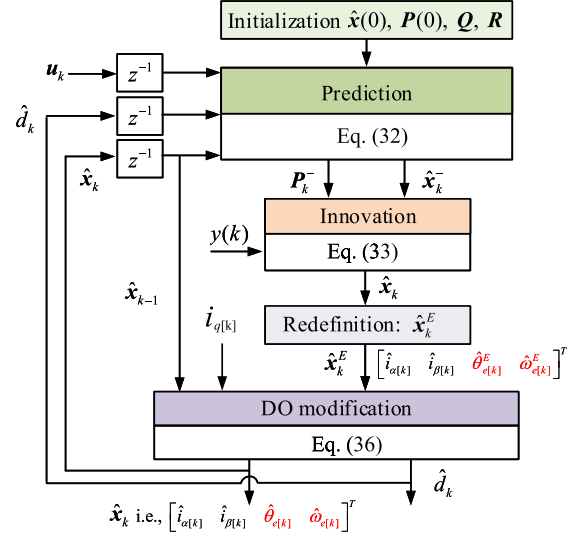


Fig. 5. Estimation fusion mechanism of the proposed CEKF.

2) *Disturbance Estimation With Linear DO*: For the sake of distinction, the states estimated by the EKF in this section are redefined as $\hat{\mathbf{x}}^E = [\hat{i}_\alpha \ \hat{i}_\beta \ \hat{\omega}_e^E \ \hat{\theta}_e^E]^T$. Under the fact that $i_q = i_\beta \cos \theta_e - i_\alpha \sin \theta_e$, it follows from (24) that the model of electrical angular velocity from EKF is

$$\begin{cases} \dot{\hat{\theta}}_e^E = \hat{\omega}_e^E \\ \dot{\hat{\omega}}_e^E = K_T i_q - \frac{n_p}{J} d(t) - \frac{B}{J} \hat{\omega}_e^E \end{cases} \quad (34)$$

where $d(t) = T_L$ is considered as the disturbance arising from external load torque T_L . Then, the linear DO can be designed as

$$\begin{cases} \dot{\hat{\theta}}_e = \hat{\omega}_e + \beta_0 (\hat{\theta}_e^E - \hat{\theta}_e) \\ \dot{\hat{\omega}}_e = K_T i_q - \frac{n_p}{J} \hat{d}(t) - \frac{B}{J} \hat{\omega}_e + \beta_1 (\hat{\theta}_e^E - \hat{\theta}_e) \\ \dot{\hat{d}}(t) = \beta_2 (\hat{\theta}_e^E - \hat{\theta}_e) \end{cases} \quad (35)$$

where $\hat{\theta}_e$ and $\hat{\omega}_e$ are the estimation of $\hat{\theta}_e^E$ and $\hat{\omega}_e^E$ and $\hat{d}(t)$ is the estimation of T_L , respectively. $\beta_i > 0$, for $i = 0, 1, 2$ are the DO gains. Then, the estimation error will converge to zero asymptotically.

In discrete-time domain, the DO (35) can be rewritten as

$$\begin{cases} e_0[k-1] = \hat{\theta}_{e[k-1]}^E - \hat{\theta}_{e[k-1]} \\ \hat{\theta}_{e[k]} = \hat{\theta}_{e[k-1]} + T_s \hat{\omega}_e(k-1) + T_s \beta_0 e_0[k-1] \\ \hat{\omega}_{e[k]} = \hat{\omega}_{e[k-1]} + T_s \left[K_T i_{q[k]} - \frac{n_p}{J} \hat{d}_{[k-1]} - \frac{B}{J} \hat{\omega}_{e[k-1]} \right] \\ \quad + T_s \beta_1 e_0[k-1] \\ \hat{d}_{[k]} = \hat{d}_{[k-1]} + T_s \beta_2 e_0[k-1]. \end{cases} \quad (36)$$

3) *Estimation Fusion*: The estimation fusion mechanism of CEKF is illustrated in Fig. 5. The embedded DO simultaneously estimates the disturbance $\hat{d}_k$ and modifies the state variables $\hat{\mathbf{x}}^E$, and then feeds them back to the first step for states estimation at the next moment.

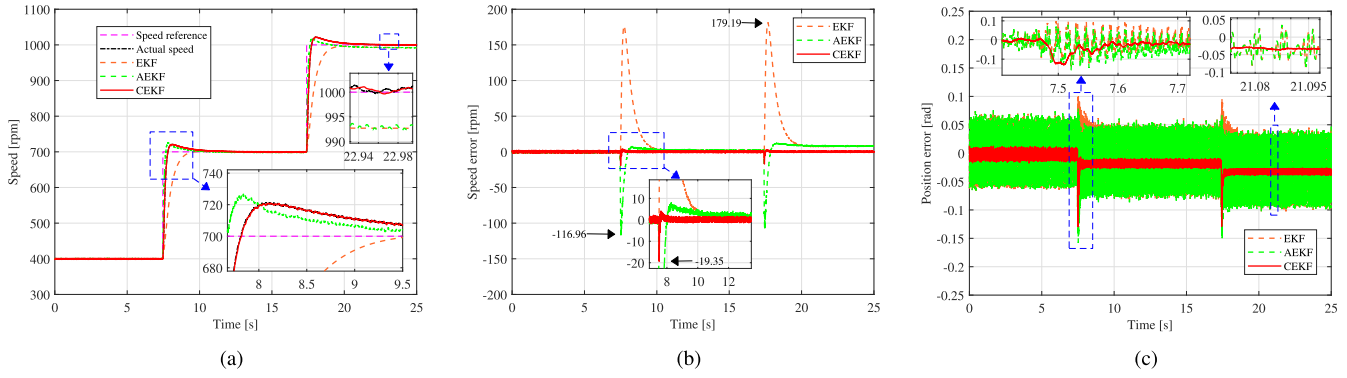


Fig. 8. Comparative results of conventional EKF, AEKF, and CEKF under step speed reference changes. (a) Speed response curves. (b) Speed estimation errors. (c) Position estimation errors.

TABLE I
KEY PARAMETERS OF SPMSM

Parameters	Units	Values
Rated power P	kW	1.5
Rated speed N_r	rpm	1500
Rated load T_L	N·m	5
Rated voltage U_r	V	380
Rated current I_r	A	4
Pole pairs N_p	*	4
Stator resistance R_s	Ω	2.16
Line-line winding inductance L_s	mH	18.7
Rotational inertia J	Kg·m ²	0.00188
PM flux linkage ψ_f	Wb	0.32
Viscous friction coefficient B	N·m·s/rad	4.1×10^{-5}

EKF, AEKF, and CEKF are kept as consistent as possible. For these matrices, it is a common practice to use diagonal matrices [23], and then normally they are tuned by a trial-and-error manner [24]. Thus, these matrices are selected as follows:

$$\mathbf{P}_{0\text{EKF}} = \mathbf{P}_{0\text{CEKF}} = \text{diag}([0.1, 0.1, 100, 5])$$

$$\mathbf{P}_{0\text{AEKF}} = \text{diag}([0.1, 0.1, 100, 5, 0.05])$$

$$\mathbf{Q}_{\text{EKF}} = \mathbf{Q}_{\text{CEKF}} = \text{diag}([0.1, 0.1, 1, 5])$$

$$\mathbf{Q}_{\text{AEKF}} = \text{diag}([0.1, 0.1, 1, 5, 0.01])$$

$$\mathbf{R} = \text{diag}([10, 10]).$$

where $\text{diag}(\cdot)$ denotes a diagonal matrix and $\mathbf{P}_0$ is initial value of the covariance matrix $\mathbf{P}$.

A. Comparison Under Dynamic Speed Regulation

Fig. 8 shows the experimental results of speed estimation, speed estimation error, and position estimation error for the three sensorless control schemes under step speed reference change. The speed reference follows a stepwise transition of 400–700–1000 rpm. As shown in Fig. 8(a), CEKF tracks the actual speed more accurately than conventional EKF and AEKF across all operating speeds. In Fig. 8(b), the maximum speed estimation errors for EKF, AEKF, and CEKF are 179.19, 116.96, and

19.35 rpm, respectively, demonstrating CEKF's superior estimation accuracy. Fig. 8(c) further confirms that CEKF maintains the smallest position estimation error throughout the process, while AEKF achieves slightly better position accuracy than EKF.

B. Comparison Under Load Disturbance

Fig. 9 presents the experimental results of conventional EKF, AEKF, and CEKF at 600 rpm under a step load disturbance. From top to bottom, the figure displays speed estimation, speed estimation error, position estimation error, and q -axis current. A sudden 1.5 N·m load is applied and later removed. In particular, before the external load is introduced, there is a dragging torque of approximately 0.40 N·m in the SPMSM system at 600 r/m, which is mainly caused by the electromagnetic drag from the unpowered asynchronous motor. After adding the load, EKF and AEKF exhibit mean speed estimation errors of 61.8 and 59.7 r/m, respectively, while CEKF maintains a significantly lower maximum speed estimation error of 6.5 r/m. Similarly, the maximum position estimation errors for EKF, AEKF, and CEKF are 0.122, 0.106, and 0.047 rd, respectively. These results demonstrate that CEKF achieves the smallest speed and position estimation errors under load disturbances, highlighting its superior disturbance rejection capability. In addition, AEKF slightly outperforms EKF in estimation accuracy.

Fig. 10 illustrates the load estimation results using the separate DO, AEKF, and CEKF. Since the actual loading process does not satisfy Assumption 2, AEKF experiences load disturbance estimation errors. These errors, along with parameter uncertainties and unknown torques (e.g., friction and drag torque), contribute to AEKF's speed estimation inaccuracies under load conditions. Compared to the separate DO, CEKF exhibits reduced estimation fluctuations, benefiting from the EKF-based prefiltering mechanism integrated within CEKF. In addition, the DO itself is characterized by low-pass filtering, resulting in smaller harmonics in the position estimation error and speed estimation error.

C. Comparison Under Parameter Mismatch

Fig. 11 presents the experimental results of conventional EKF, AEKF, and CEKF under no-load condition with parameter

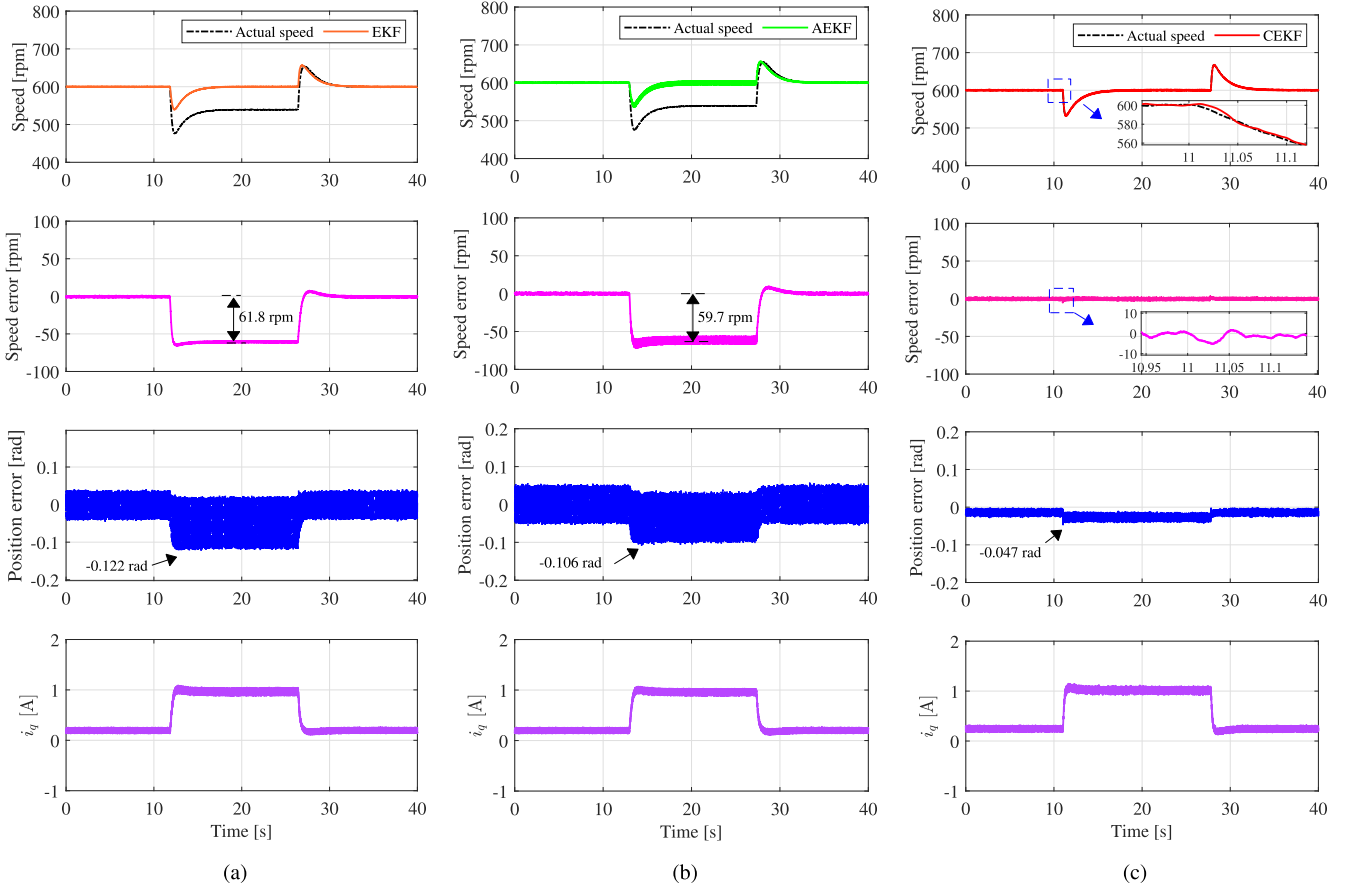


Fig. 9. Comparison of speed estimation (and error), position estimation error, and q -axis current under step load changes. (a) EKF. (b) AEKF. (c) CEKF.

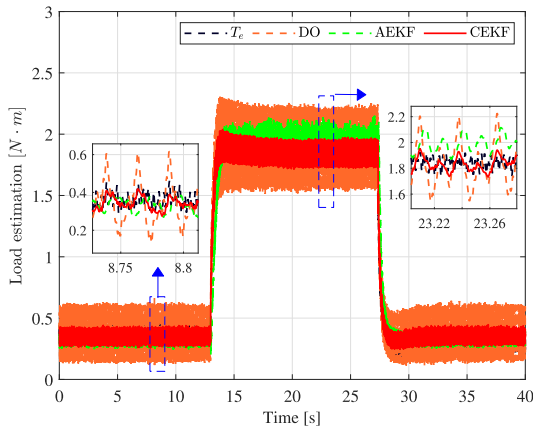


Fig. 10. Comparative results of load disturbance estimation.

mismatch at 600 r/m. Fig. 11(a) illustrates the results under resistance (R_s) mismatch, showing both speed and position estimation errors for all three methods. The cases $R_s = 0.5\bar{R}_s$ and $R_s = 1.5\bar{R}_s$ represent stator resistance set to 50% and 150% of the nominal value $\bar{R}_s$, respectively. As observed, EKF and AEKF exhibit significant speed estimation fluctuations under $\pm 50\%$ resistance variations, whereas position estimation errors remain largely unchanged. Fig. 11(b) illustrates the results under

inductance (L_s) mismatch at 600 r/m, displaying speed and position estimation errors for all three methods. Similarly, $L_s = 0.5\bar{L}_s$ and $L_s = 1.5\bar{L}_s$ correspond to 50% and 150% of the nominal inductance $\bar{L}_s$. Both EKF and AEKF show substantial fluctuations in speed and position estimation errors under $\pm 50\%$ inductance variations, indicating high sensitivity to inductance changes. Fig. 11(c) shows the position and speed estimation error of these three methods under PM flux linkage ψ_f mismatch. The cases $\psi_f = 0.8\bar{\psi}_f$ and $\psi_f = 1.2\bar{\psi}_f$ denote PM flux linkage set to 80% and 120% of the nominal value $\bar{\psi}_f$, respectively. As shown, the variations in flux linkage ψ_f significantly affect the speed estimation errors of EKF and AEKF, which in return impacts their position estimation errors. Similar phenomenon also exists and has been discussed in [25] and [26]. Fig. 11(d) shows the position and speed estimation error of these three sensorless methods with rotational inertia mismatch. The cases $J = 0.5\bar{J}$ and $J = 1.5\bar{J}$ denote rotational inertia set to 50% and 150% of the nominal value $\bar{J}$, respectively. As observed, the position and speed estimation errors of these three sensorless algorithms remain almost unchanged for variations in rotational inertia. Besides, a quantitative analysis is also provided in Fig. 11(e). The mean absolute error (MAE) is defined as

$$\text{MAE} = \frac{1}{N} \sum_{k=1}^N |e(k)| \quad (40)$$

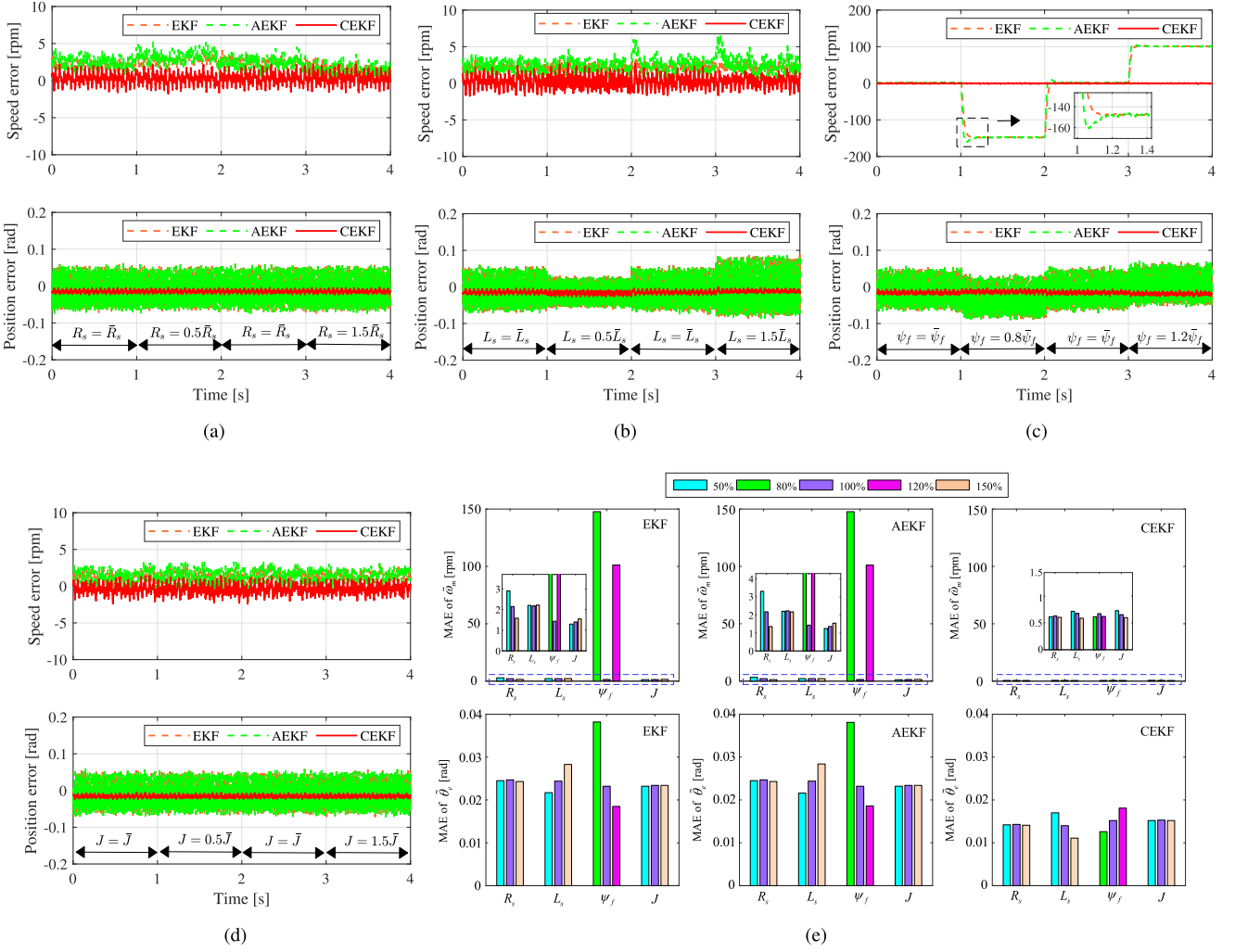


Fig. 11. Comparative results of EKF, AEKF, and CEKF under parameter mismatch. (a) Resistance mismatch. (b) Inductance mismatch. (c) PM flux linkage mismatch. (d) Rotational inertia mismatch. (e) MAE for speed estimation error and position estimation error.

where e is the position estimation error $\tilde{\theta}_e$ or speed estimation error $\tilde{\omega}_m$, k is the sampling instant, and N is the number of sampling points. Here, N is selected as 4000.

In contrast, as shown in Fig. 11(e), CEKF maintains stable speed and position estimation across resistance, inductance, flux linkage, and rotational inertia variations, demonstrating its superior robustness against parameter mismatches. In addition, AEKF exhibits slightly higher sensitivity to parameter variations than EKF.

D. Computational Cost

Computational cost of the three estimation methods is compared based on execution times, i.e., the average execution time, the maximal execution time, as well as the minimal execution time. The dSPACE MicroLabBox has dual-core real-time processor at 2 GHz, which shows powerful computing capabilities. The execution times of these three algorithms are summarized in Table II. AEKF exhibits the highest computational cost, primarily due to the increased dimensionality of its state vector

TABLE II
COMPUTATIONAL COST OF THESE SENSORLESS CONTROL SCHEMES

Algorithm	Average	Maximal	Minimal
EKF	18.8 μ s	20.5 μ s	16.2 μ s
AEKF	29.7 μ s	35.1 μ s	27.8 μ s
CEKF	22.6 μ s	29.4 μ s	20.3 μ s

and covariance matrix. EKF demonstrates the fastest performance, whereas the proposed CEKF achieves a balance between accuracy and efficiency.

VI. CONCLUSION

This article addresses the challenges of sensorless PMSM control under unknown disturbances. Conventional approaches, which separately estimate speed/position and disturbances, often suffer from instability, especially when disturbances

significantly degrade estimation accuracy. To overcome this limitation, we propose a novel estimation-control framework that simultaneously estimates system states and disturbances. Specifically, we introduce a CEKF based on a DO. In addition, we compare the proposed CEKF with the AEKF and the conventional EKF. Extensive hardware experiments under varying speed regulations, unknown load disturbances, and parameter mismatches validate the feasibility and effectiveness of our approach. Comparative results show that the proposed simultaneous estimation framework significantly enhances speed and position estimation accuracy over conventional methods. This improvement leads to enhanced system stability and robustness, making our framework a promising solution for high-performance sensorless PMSM control. Future work will focus on suppressing non-Gaussian noise and estimating key time-varying parameters to further enhance system performance.

APPENDIX

JACOBIAN MATRICES FOR CEKF AND AEKF

Define the Jacobian matrices in the CEKF

$$F(x) = \frac{\partial f}{\partial x} \Big|_{x=x(t)} = \begin{bmatrix} -\frac{R_s}{L_s} & 0 & \frac{\psi_f \sin \theta_e}{L_s} & \frac{\psi_f \cos \theta_e}{L_s} \\ 0 & -\frac{R_s}{L_s} & -\frac{\psi_f \cos \theta_e}{L_s} & -\frac{\psi_f \sin \theta_e}{L_s} \\ K_T \sin \theta_e & K_T \cos \theta_e & -\frac{B}{J} & -K_T \omega_e (i_\beta \sin \theta_e + i_\alpha \cos \theta_e) \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H(x) = \frac{\partial h}{\partial x} \Big|_{x=x(t)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}. \quad (41)$$

Define the augmented Jacobian matrices in the AEKF

$$\bar{F}(\bar{x}) = \frac{\partial \bar{f}}{\partial \bar{x}} \Big|_{\bar{x}=\bar{x}(t)} = \begin{bmatrix} F(x)_{4 \times 4} & \Gamma_{4 \times 1} \\ \mathbf{0}_{1 \times 4} & \mathbf{0}_{1 \times 1} \end{bmatrix}_{5 \times 5}$$

$$\bar{H}(\bar{x}) = \frac{\partial \bar{h}}{\partial \bar{x}} \Big|_{\bar{x}=\bar{x}(t)} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad (42)$$

where $\Gamma_{4 \times 1} = [0 \ 0 \ -\frac{n_p}{J} \ 0]^T$.

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